LOW-END INTEL MOTHERBOARDS

Intel	Jetway	Maxforce	Mercury	MSI	Gigabyte	ASRock	ASUS
D845GVSR	IGVM	MS-6714	Pi845GLM-L	MS-845GVML	GA-8I865GVMK	M266A	P4V533-MX-UAYZ
VIA KM400	Intel 845GV	Intel 845GV	Intel 845GL	Intel 845GV	Intel 865GV	VIA P4M266A	VIA P4M266A
Socket A	Socket 478	Socket478	Socket 478	Socket 478	Socket 478	Socket 478	Socket 478
P4 3.06 / 533	P4 3.06 / 533	P4 3.06 / 533	P4 3.06 / 533	P4 3.06 / 533	P4 3.2 / 800	P4 3.06 / 533	P4 3.06 / 533
DDR 333 / 2	DDR 266 / 2	DDR 333 / 2	DDR 266 / 2	DDR 333 / 2	DDR 400 / 4	DDR 266 and SD PC133 / 4	DDR 266 / 2
Micro ATX	Micro ATX	Micro ATX	Micro ATX	Micro ATX	Micro ATX	Micro ATX	Micro ATX
✘/✔	✘/✔	✘/✔	✘/✔	✘/✔	✘/✔	✔/✔	✔/✔
3	3	3	3	3	3	3	3
✔	✔	✔	✔	✔	✔	✔	✔
6/4/2	6/2/4	6/4/2	6/4/2	6/4/2	8/4/4	6/4/2	6/4/2
✘	✘	✔	✔	✔	✔	✔	✔
✘	✘	✘	✘	✔	✔	✘	✔
✘	✘	✔	✘	✘	✔	✘	✔
✘	✘	✔	✘	✘	✔	✘	✔
3	3	2.5	3	2	2.75	3	2.75
✔ / ✔	✔ / ✔	✔ / ✔	✔ / ✔	✔ / ✔	✔ / ✔	✔ / ✔	✔ / ✔
✔	✔	✔	✔	✔	✔	✔	✔
17.37	29.77	23.87	27.70	26.77	39.93	19.40	18.30
53.43	62.27	69.07	69.50	68.80	103.40	21.67	21.90
Failed	23.2	16.5	22.6	23.3	27.4	17	17.3
12.8	18.9	15.9	18.8	18.3	21.9	13	12.6
115	100	124	103	101	71	154	153
1,324	1,476	1,524	1,527	1,534	3,006	468	400
3,151	3,669	3,567	3,574	3,556	4,911	3,950	3,876
2,084	2,374	2,345	2,384	2,659	4,376	1,572	1,603
526	455	460	479	488	578	186	191
3,559	3,363	3,290	3,317	3,376	2,903	3,275	3,222
2,500	2,781	2,456	2,774	2,801	3,936	2,401	2,306
7,171 / 2,036	6,992 / 2,071	6,500 / 2,057	7,297 / 2,156	7,077 / ,2058	10,044 / 4,024	9,289 / 3,621	9,034 / 3,807
17,257/20,417	17,373/20,592	17,284/20,303	17,276/20,461	17,290/20,487	24,605/34,773	23211/33834	23,300/32,769
1,665	1,553	1,544	1,644	1,538	4,219	776	808
1,668	1,556	1,546	1,643	1,541	4,223	776	808
29,217	27,534	26,304	24,353	26,578	24,570	27,355	27,453
32.21	47.05	41.25	46.86	47.11	65.27	33.15	32.75
8.80	7.80	11.50	8.80	10.20	14.65	9.80	12.65
41.01	54.85	52.75	55.66	57.31	79.92	42.95	45.40
10.25	18.28	16.48	21.20	14.33	14.53	13.01	12.97
4,000	3,000	3,200	2,625	4,000	5,500	3,300	3,500
C+	B+	B+	A-	B	B	B-	B-

supply connector and the 12 V auxiliary power points on the board. If these connectors are on the other side of the cooling fan, the wires may get entangled with its blades, thus disrupting proper air ventilation near the CPU. Choose boards that have these power points on the extreme right-hand side away from the fan.

The IDE and floppy drive connectors can be a nuisance if they are too close to each other. Choose a board where these connectors are not placed linearly, but next to each other. In the new breed of boards that come with dual-channel memory, the RAM slots are too close to each other. Ensure that there is enough spacing between them. RAM modules with a heat spreader require adequate space for proper air circulation to dissipate the heat.

I/O sub-system test. The gaming tests were run to gauge the difference between the Intel Extreme graphics integrated on the 845GV-based boards, and the S3 graphics used by the others. All the boards managed to run most of our gaming tests, but the gameplay was jerky, and at times unplayable. One interesting point is the difference in image quality between the two graph-

The Intel 865GV-based Gigabyte GA-81865GVMK has good features and performed well